

PALAKURU SAI

Python Developer | Data Analyst | Software Engineer

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PROFESSIONAL SUMMARY

ECE graduate and PCEP-30-02 certified Python Developer with hands-on experience in Python, SQL, Pandas, NumPy, Matplotlib, data analysis. Built a Marine Debris Detection system using deep learning and a full-stack Personal Finance Dashboard with interactive data visualizations. Strong foundation in OOP, Data Structures & Algorithms, REST APIs, and version control (Git/GitHub). Seeking SDE, Python Developer, or Data Analyst roles (full-time/internship, remote or on-site, across India).

TECHNICAL SKILLS

- Languages: Python, JavaScript, HTML5, CSS3, SQL
- Frameworks & Libraries: NumPy, Pandas, Matplotlib
- CS Fundamentals: OOP, Data Structures & Algorithms, Operating Systems, Computer Networks
- Tools & Databases: Git, GitHub, VS Code, MySQL, Excel (VLOOKUP, Pivot Tables)

PROJECTS

Marine Debris Detection Using Side Scan Sonar | Python, YOLOv8, OpenCV, Pandas, Matplotlib | [GitHub](#)

- Engineered an object detection pipeline to identify marine debris in side-scan sonar images using YOLOv8, improving manual review efficiency.
- Preprocessed and augmented sonar image datasets with noise reduction and contrast enhancement techniques to improve model input quality.
- Trained and evaluated the model on domain-specific data, achieving strong detection accuracy and validating results via precision/recall metrics.

Personal Finance Dashboard | Python, HTML, CSS, JavaScript, Git | [GitHub](#)

- Developed a full-stack web application to track income, expenses, savings, and financial trends with real-time updates.
- Designed interactive data visualizations and automated data-processing pipelines to summarize spending patterns.
- Built a responsive, mobile-friendly UI to improve usability across devices.

IPL Data Analyzer | Python, Pandas, NumPy, Matplotlib, SQL, Git | [GitHub](#)

- Analyzed 15+ seasons of IPL data using Python, Pandas, and SQL to uncover team and player performance trends.
- Processed and analyzed 200,000+ delivery records and 1,000+ match records using Pandas and NumPy.
- Developed 15+ visualizations and analytical reports using Matplotlib to present cricket performance insights.

EXPERIENCE & INTERNSHIPS

Virtual Python Developer Intern — Prowork Foundation 2024

- Developed Python applications including a weather app, a data analysis dashboard, and automation scripts to streamline repetitive tasks.
- Strengthened programming fundamentals, debugging, and problem-solving skills in a professional virtual team environment.

Competitive Programming & Advanced Algorithmic Techniques May – Jul 2025

- Solved complex DSA problems under time constraints, improving algorithmic thinking and coding speed.
- Strengthened understanding of recursion, dynamic programming, and graph-based problem-solving techniques.

EDUCATION

- B.Tech, Electronics & Communication Engineering — SVCE, Tirupati | CGPA: 7.79 | 2022 – 2026
- Intermediate (MPC) — Narayana Junior College | 7.45 | 2020 – 2022
- SSC — Narayana EM School | 10.0/10.0 | 2019 – 2020

CERTIFICATIONS

- PCEP-30-02 — Certified Entry-Level Python Programmer
- Python Essential Training — LinkedIn Learning
- SQL (Intermediate) & Python (Basic) — HackerRank

ACHIEVEMENTS

- Built and deployed two end-to-end Python projects independently.
- Earned PCEP-30-02 certification, validating strong core Python programming fundamentals.
- Consistently solved advanced DSA problems (recursion, dynamic programming, graphs) under timed constraints.